

Capstone Project: Forcaster

Project Overview

Introduction and Problem Statement

The capstone project serves as a culminating experience, allowing you to apply the knowledge and skills you have acquired throughout your studies to a real-world problem or challenge. This project applies to an environment where knowledge and the subsequent prediction of events are critical. The agentic system here is more focused on collecting knowledge through channels such as news, markets, and weather. The potential applications here include systems to analyze market conditions for opportune financial trades, to accumulate weather-related data to provide weather predictions, or to monitor news sources to forecast downstream effects. These are just a couple of examples, but you can develop any forecasting system that relies primarily on the collection of knowledge for question-answer responses for a specific problem you want to solve.

Project Goals and Objectives

The primary goal of this capstone is to demonstrate your ability to use the agentic tools and concepts acquired throughout the course.

Concepts	Description	Success Metric
Tool Calling	Utilize an agentic architecture to provide the large language model with tool-calling abilities.	The agent can execute tool calls and use functions while understanding their outputs.
Reasoning Architectures and Chain of Thought	Apply a basic reasoning loop, such as ReAct, so the agent can interact with its current environment.	The agentic system demonstrates the ability to reason and recover from missteps.
Knowledge and Memory	Utilize a retrieval-based memory mechanism such as RAG or other proposed	The agentic system can recall past task executions or use a store of knowledge, such as

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	methods to retain knowledge between tool calls.	API documentation or other data.
Further Reasoning	Further build up the reasoning loop using concepts such as Tree of Thought	The system can more clearly outline the agent's reasoning architecture and thought process.
Multi-agent Coordination	Develop the agentic system to leverage parallelism and specialized agents to assist with various tasks.	Demonstration of at least one agent to execute tasks.
Safety	Ensure that the agentic system is safe to deploy and that specified guardrails are in place to prevent unintended task execution.	Statement of potential unintended actions the system may execute and the steps taken to address this.

Deliverables

The final project deliverables will include:

- **Final Report:** A comprehensive document detailing the problem, methodology, results, and conclusions.
- **Presentation:** A formal presentation summarizing the project for the faculty and peers.